

Title: Spam Email Detection Using Machine Learning Name: [Your Name] Branch: B.Tech CSE (AI)

Introduction Email spam is unwanted messages. Machine learning helps detect spam automatically.

Objective Classify emails as Spam (1) or Ham (0).

Technologies Used Python, Pandas, Scikit-learn, Matplotlib, CountVectorizer

Dataset Emails_data.csv with message and label columns.

Data Preprocessing Convert labels and text into numeric format.

Feature Extraction Use CountVectorizer to convert text to numbers.

Visualization Scatter plot of word frequency vs label.

Train-Test Split 80% training, 20% testing.

Model Used Multinomial Naive Bayes.

Training Model learns spam patterns.

Prediction Predict spam or not spam.

Accuracy Measure performance using accuracy score.

User Input User enters email text for prediction.

Advantages Fast, automatic, improves security.

Limitations Depends on dataset quality.

Future Scope Use deep learning, GUI.

Conclusion Spam detection system built successfully.

Thank You Any Questions?